

QA AUTOMATION TEST STRATEGY

ImportActiveComponent - Comprehensive Test Scenario Map

Playwright + APIRequestContext + Excel contract checks with SheetJS

Field	Value
Route	/dashboard/bienelectrico/importar
UI context	Dashboard panel, 600 px wide
Backend load type	1 description=importacion authenticated login attached
Evidence reviewed	Current Angular component/services, supplied XLSX, test-id reference, edit/create QA maps, Playwright suite

Recommendation: Automate the import as a state-changing electrical-asset workflow, not as a generic uploader. Keep deep end-to-end coverage on template compatibility, validation, final POST behavior, and post-import GET reconciliation. Give History and error-CSV export only one stable contract test each because their business value is low.

1. What the evidence says

Area	Observed contract
Workbook	3 sheets: Plantilla, Instrucciones, Codigos Dane; 102 headers from A to CX.
Required fields	7 dark-header columns: ARTICULO, N° PLACA, FOTOS (enlace), PLANILLA (ARCGIS), DEPARTAMENTO, MUNICIPIO, COD_LOCALIZACION_DANE.
Optional fields	95 light-green columns. Header comments carry type and maximum-length guidance.
Location rules	Departamento and Municipio use dependent validation lists through row 5000. Instructions say DANE codes are persisted.
Formatting rules	Decimal point, dates DD/MM/AAAA, and complete https:// URLs are explicitly documented in the workbook.
Frontend gate	.xlsx/.xls, one file, maximum 10 MB; successful backend validation is required before import.
Error handling	HTTP 400, 402, and 422 open the detailed validation-error path; other statuses use snackbar/transient error handling.
Current locator risk	The supplied PDF lists 9 data-testid values, but the inspected HTML snapshot contains none. Confirm the target branch before writing locators.

2. Automation principles

- Use the downloaded backend template as the source of truth. A version-controlled header snapshot should fail loudly when the backend changes the column contract.
- Acquire available plates dynamically from GET /electrical-assets/article-resume/available-plates. Do not hard-code plates that other tests can consume.
- Treat each successful import as destructive test data. Run state-changing import suites serially and reserve disposable QA plates.
- Assert both network and persistence: validate POST, final load POST, then GET /electrical-assets/article-resume/{plate} for the database-facing result.
- Separate deterministic UI tests with mocked responses from a small number of live backend contract/E2E tests.
- For update and blank-clearing behavior, restore the original record in finally when the API supports a reliable PATCH restore; otherwise use resettable seeded data only.

3. Recommended execution portfolio

Cadence	Scope	Data rule
Pull request	P0 local gates and mocked API/UI states; template header contract; no bulk state consumption.	Fast and deterministic
Daily smoke	One disposable valid plate: validate, import, post-import GET, cleanup/reset.	Serial worker
Nightly	Existing-plate replacement, blank clearing, 25-50 valid plates, atomicity, retry, pagination.	Dedicated seed pool
Release	10 MB boundary, .xls compatibility, 4999-row workbook boundary, concurrency and performance.	Controlled environment

4. Comprehensive scenario matrix

Priority is risk-based: P0 protects creation/update integrity, P1 protects validation and recovery, and P2 covers secondary UX. A scenario marked Contract can run below the UI; UI+API means Playwright drives the component and captures/reconciles network behavior.

4.1 Template, access, and local uploader gates

ID	Pri	Layer	Scenario	Test data	Key assertions	Cadence
QA-IMP-001	P0	UI	Route and authentication gate	Authenticated and unauthenticated sessions	Authenticated user can open the 600 px panel. Unauthenticated/expired session is redirected or denied. Import tab is default and submit starts disabled.	PR
QA-IMP-002	P0	UI+API	Download current template	Backend template blob	Exactly one GET /import/template?loadType=1; response is non-empty Excel; downloaded name matches plantilla_importacion_YYYY-MM-DD_HH.mm.xlsx.	PR
QA-IMP-003	P0	Contract	SheetJS workbook/header contract	Downloaded template	XLSX.read succeeds; sheets are Plantilla/Instrucciones/Codigos Dane; first row has exactly 102 unique headers in approved order from A to CX.	PR
QA-IMP-004	P0	Contract	Required-column contract	Downloaded template	Approved required set is exactly A,B,C,D,E,F,Z. Comments/types remain available. Flag a backend template change instead of silently adapting the test.	PR
QA-IMP-005	P1	UI+API	Accept valid .xlsx	valid_minimal_1.xlsx	File is readable, validation POST occurs once, uploader reaches completed, and Importar activo becomes enabled only after validation success.	PR
QA-IMP-006	P1	UI+API	Accept real legacy .xls	valid_minimal_1.xls	A genuine BIFF .xls is accepted and validated. Do not satisfy this with an .xlsx merely renamed .xls.	Release
QA-IMP-007	P1	UI	Reject local file misuse	pdf/csv/txt, renamed file, unreadable UploadedFile, two-file drop	Unsupported extension, unreadable file, and multiple-file attempts never call validation; uploader shows the correct local error and submit remains disabled.	PR
QA-IMP-008	P1	UI	10 MB boundary	10 MB exactly; 10 MB + 1 byte	A file at the limit reaches backend validation; one byte over is rejected locally as Archivo muy grande; no validation request for the oversized file.	Release

4.2 Backend validation and spreadsheet business rules

ID	Pri	Layer	Scenario	Test data	Key assertions	Cadence
QA-IMP-009	P0	UI+API	Header-only or empty workbook	header_only.xlsx; truly blank workbook	Backend response with totalRows=0 yields El archivo no contiene datos, uploader Sin datos, null validatedFile, and disabled submit.	PR
QA-IMP-010	P1	Contract	Sparse rows	Rows 2 and 4 populated; rows 3/5 blank	Blank lines are ignored consistently; totalRows counts actual asset rows; error row numbers still point to physical Excel rows.	Nightly
QA-IMP-011	P0	Contract	Each required cell empty	7 generated files, one missing required value per file	Validation rejects ARTICULO, N° PLACA, FOTOS, PLANILLA, DEPARTAMENTO, MUNICIPIO, and COD_LOCALIZACION_DANE independently with row, column, description.	PR
QA-IMP-012	P0	Contract	Header mutation matrix	missing, renamed, duplicate, reordered, extra, whitespace header	Backend behavior is explicit for every mutation. Import cannot proceed on a structurally incompatible workbook. Reordering is asserted according to the agreed backend contract, not guessed.	PR
QA-IMP-013	P0	API+UI	Valid available plate	Plate reserved from available-plates API	Validation succeeds only when the plate/article relationship is eligible. The UI enables import and preserves the exact File object for final load.	Daily
QA-IMP-014	P0	Contract	Unknown or unavailable plate	Unknown, baja/devolución, already consumed plate	Validation rejects ineligible plates with an actionable row/column error; no final load request is possible.	Nightly
QA-IMP-015	P0	Contract	Duplicate plate within workbook	Same plate twice; conflicting values	Validation reports both the duplication and relevant row numbers. No partial import or ambiguous last-row-wins behavior occurs unless explicitly specified.	Nightly
QA-IMP-016	P0	Contract	Location relationship	Valid/invalid department, municipality and DANE combinations	Department and municipality are valid codes, municipality belongs to department, and COD_LOCALIZACION_DANE is consistent. Name-only and code-name inputs follow workbook instructions.	Nightly
QA-IMP-017	P1	Contract	Types, lengths, and formats	Boundary matrix from header comments	For representative text/number/decimal/date/URL fields: max is accepted, max+1 rejected; comma decimals, impossible dates, malformed URLs, negative/decimal values are handled by documented backend rules.	Nightly
QA-IMP-018	P1	Contract	Whitespace, Unicode, and formulas	Trimmed values, accents, emoji, formulas, numeric-looking text	Leading zeros in plate are preserved; whitespace normalization is deterministic; formulas are rejected or evaluated per contract; Spanish accents survive end to end.	Nightly
QA-IMP-019	P1	UI+API	Detailed list errors	400, 402, 422 with errors/data/data.errors/array/message	Uploader shows Datos inválidos; summary alert opens; adapters display Spanish or English keys; dialog preserves row/column/description; import stays disabled.	PR
QA-IMP-020	P1	UI+API	Non-list validation failures and recovery	401, 403, 413, 500, timeout	Snackbar/transient message is visible; retry triggers a new validation; cancel/remove clears every validation/import signal; a prior success cannot enable a newly failed file.	PR

4.3 Final import, API contract, and persisted asset state

ID	Pri	Layer	Scenario	Test data	Key assertions	Cadence
A QA-IMP-021	P0	UI+API	Create one asset end to end	valid_minimal_1.xlsx + disposable available plate	Validation succeeds; one final load POST succeeds; success snackbar appears; uploader clears; GET article-resume/{plate} returns the created asset with expected required values.	Daily
QA-IMP-022	P0	API	Final POST request contract	Validated workbook	POST /import/load uses multipart field file with original filename and query params loadType=1, description=importacion, user=current login. Browser supplies multipart boundary.	PR
QA-IMP-023	P0	API	Pre/post state reconciliation	Reserved available plate	Capture precondition from available-plates; after load, poll GET article-resume/{plate}; compare normalized fields; confirm plate is no longer offered as available when business rules require it.	Daily
create QA-IMP-024	P0	API+UI	Mass import valid plates	25-50 dynamically reserved plates	Validation totalRows=N; load succeeds once; every plate is retrievable; no duplicates/missing records; history processed count equals N. Record validation/load duration without a fragile hard SLA.	Nightly
QA-IMP-025	P0	API	update and check Existing plate full replacement	update_full_existing_plate.xlsx	GET baseline, import changed values across multiple asset sections, GET again, and compare the complete expected snapshot. Prove import replacement semantics rather than assuming PATCH behavior.	Nightly
QA-IMP-026	P0	API	Blank optional cells clear values	update_clear_optional.xlsx	Start with non-empty optional values, leave selected cells blank, import, then GET. Assert the canonical cleared representation (null or empty string) field by field and that required identifiers remain intact.	Nightly
QA-IMP-027	P0	API	Mixed valid/invalid batch atomicity	One valid + one invalid row	Validation blocks final load. If load is forced at API level, assert the agreed all-or-nothing behavior and verify the valid row was not partially persisted.	Nightly
QA-IMP-028	P0	UI+API	duplicate import Double submit and re-import	Validated workbook; rapid click; same workbook twice	Only one load request from rapid clicks. Re-importing the same plate/workbook does not create duplicates; response follows the documented update/idempotency contract.	PR
No next? QA-IMP-029	P1	UI+API	Load failure then retry	500/timeout on first load; success second	Validated state is retained after final-load failure; message is visible; retry sends the same file once; success clears uploader. Validation is not repeated unless product requires it.	PR
QA-IMP-030	P1	API	Concurrent imports	Two workbooks sharing one plate; disjoint control	Same-plate conflict has one deterministic winner/rejection and no corrupt merge. Disjoint imports remain isolated. Run only with resettable fixtures.	Release
QA-IMP-031	P1	API	Authenticated user attribution	Known QA login	Final POST user matches authenticated login and the resulting load/history record attributes the operation consistently. Empty/stale login is rejected or surfaced.	Nightly

4.4 History, error export, layout, and UX

ID	Pri	Layer	Scenario	Test data	Key assertions	Cadence
QA-IMP-032	P2	UI+API	History request contract	Mocked page 0	Selecting Historial sends GET /electrical-assets/loads?page=0&size=5&sort=createdAt,desc; newest records render first with date, status, records, description.	PR
QA-IMP-033	P2	UI	Infinite scroll pagination	3 mocked pages; overlapping IDs	Near-bottom scroll loads one next page at a time; loading guard prevents duplicates; last=true stops requests; repeated IDs follow the agreed rendering rule.	Nightly
QA-IMP-034	P2	UI	History empty state	Empty array and empty paginated response	No hay registros para mostrar is visible and no endless loader remains.	PR
QA-IMP-035	P2	UI	History/template failure feedback	500/timeout	Document current gap: failures only stop loading/log to console. Keep one regression test pending until visible feedback is implemented.	Backlog
QA-IMP-036	P2	UI+API	History after successful import	Successful load then tab switch	Document current behavior: history is not refreshed immediately but is reloaded when Historial is selected. Assert the actual product decision once confirmed.	Nightly
QA-IMP-037	P2	Component	Validation-error CSV contract	Spanish/English error keys, separators, quotes, newlines	One contract test is enough: UTF-8 BOM, semicolon headers, RFC-style escaping, fallback values, sanitized timestamped filename. Avoid a large E2E suite for this low-value feature.	PR
QA-IMP-038	P1	UI	600 px panel layout	No file, validating, error alert, long filename, history	No horizontal clipping; actions remain reachable; error dialog fits viewport; long filenames/messages wrap; scroll is contained in intended region.	PR
QA-IMP-039	P2	UI	Keyboard and focus	Keyboard-only path	Tabs, download, uploader, cancel, submit, dialog, and close controls are reachable in logical order; focus remains visible and returns from dialog.	Release
QA-IMP-040	P1	UI	Spanish encoding regression	All visible labels/messages	No mojibake such as ValidaciÃ³n, botÃ³n, or mÃ¡s appears in DOM, snackbar, CSV headers, or downloaded filenames. Current source snapshot shows encoding risk.	PR

5. High-risk API verification recipes

5.1 Create from an available plate

- Reserve a disposable plate with GET /electrical-assets/article-resume/available-plates?search=0&page=0&size=N and filter out baja/devolución states.
- Build a minimal workbook using the exact downloaded header row. Populate all 7 required columns and valid relationships.
- Drive the UI upload and capture POST /electrical-assets/import/validate?loadTypeId=1. Assert multipart file name and successful totalRows=1.
- Click Importar activo and capture POST /electrical-assets/import/load. Assert query params and the same file payload.
- Poll GET /electrical-assets/article-resume/{plate} only for a short, bounded consistency window, then compare normalized persisted values.

5.2 Existing plate replacement and blank clearing

Data safety: This test is intentionally destructive. Use a seeded plate dedicated to this suite, run serially, and restore the baseline in finally only if PATCH restoration can reproduce null/empty values exactly.

- GET the complete baseline asset and select non-empty optional fields from different nested sections.
- Import a row that changes required and optional fields while leaving chosen optional cells blank.
- GET the asset again and compare a complete expected snapshot, not only the changed fields.
- Assert blank values against a canonical field-level expectation: null or empty string. Do not accept either interchangeably unless the API contract explicitly does.
- Verify fields omitted by the row are cleared/preserved according to full-replacement semantics. This is the main product decision that the frontend cannot reveal.

5.3 Massive import

- Use 25-50 plates for nightly QA. A 500/4999-row test is a release/performance test and needs a resettable seed pool plus a larger time budget.
- Generate plates dynamically and stop early with a clear skip if the reserved pool is insufficient. Do not silently reuse consumed plates.
- Capture validate duration, load duration, returned counts, history records, and per-plate GET results. Avoid a brittle client-side SLA until the backend team defines one.
- Run with workers=1 and tag the suite as stateful. Parallel Playwright workers can consume each other's available plates.

6. Excel fixture catalog

Fixture	Construction	Primary use
valid_minimal_1.xlsx	One available plate; exactly 7 required values	P0 create smoke
valid_batch_25.xlsx	25 dynamically reserved plates	Nightly massive import
header_only.xlsx	Approved headers, no data	Empty spreadsheet
missing_required_<column>.xlsx	Seven generated variants	Required-field matrix
headers_<mutation>.xlsx	Missing/renamed/duplicate/reordered/extra/trimmed	Structure contract
sparse_rows.xlsx	Valid rows separated by blank rows	Physical row numbering
invalid_dane_relation.xlsx	Department/municipality/DANE mismatch	Relationship rule
duplicate_plate.xlsx	Same plate twice with conflicting values	Duplicate semantics
format_boundaries.xlsx	Max/max+1, dates, decimals, URLs, Unicode	Representative field types
update_full_existing_plate.xlsx	Existing seeded plate, many changed values	Full replacement
update_clear_optional.xlsx	Existing seeded plate, selected optional cells blank	Null/empty clearing
mixed_atomicity.xlsx	One valid and one invalid row	No partial commit
size_10mb.xlsx / size_10mb_plus_1.xlsx	Deterministic padded workbooks	Local size boundary

Fixture-generation rule

Use SheetJS to read and assert the workbook contract. For generated negative fixtures, write a minimal workbook with the exact approved header row and only the cells needed by the scenario. If preserving dropdowns, comments, named ranges, and styling matters, copy the downloaded template and use a library/workflow that preserves those Excel features; do not assume a read/write round trip preserves all workbook metadata.

7. Automation skeletons

7.1 SheetJS header check

```
import fs from 'node:fs/promises';
import * as XLSX from 'xlsx';

const downloadedPath = await download.path();
expect(downloadedPath).not.toBeNull();
const buffer = await fs.readFile(downloadedPath!);
const workbook = XLSX.read(buffer, { type: 'buffer' });
expect(workbook.SheetNames).toEqual(['Plantilla', 'Instrucciones', 'Codigos Dane']);

const rows = XLSX.utils.sheet_to_json<unknown[]>(workbook.Sheets.Plantilla, {
  header: 1, raw: true, blankrows: false,
});
const headers = rows[0].map(String);
expect(headers).toEqual(EXPECTED_IMPORT_HEADERS);
expect(headers).toHaveLength(102);
expect(new Set(headers).size).toBe(102);
```

7.2 UI request capture plus persisted GET

```
const validate = page.waitForResponse(r =>
  r.url().includes('/electrical-assets/import/validate?loadTypeId=1') &&
  r.request().method() === 'POST');
await importPage.upload(filePath);
expect((await validate).ok()).toBeTruthy();

const load = page.waitForResponse(r =>
  r.url().includes('/electrical-assets/import/load') &&
  r.request().method() === 'POST');
await importPage.submit.click();
const loadResponse = await load;
expect(loadResponse.ok()).toBeTruthy();
const request = loadResponse.request();
expect(new URL(request.url()).searchParams.get('loadTypeId')).toBe('1');

await expect.poll(async () => {
  const persisted = await apiContext.get(`/electrical-assets/article-resume/${plate}`, {
    headers: { Authorization: `Bearer ${authToken}` },
  });
  return persisted.status();
}).toBe(200);
```

8. Locator contract and missing test IDs

The supplied test-id reference exposes the following stable anchors. Use them when they exist in the target build, but first resolve the mismatch with the inspected HTML snapshot.

Available data-testid	Purpose
importActiveTabsContainer	Import/History tabs container
importActiveImportTab	Import tab
importActiveHistoryTab	History tab
importActiveInstructionsContainer	Steps 01-03
importActiveDownloadTemplateButton	Download template
importActiveUploaderContainer	Uploader
importActiveActionsContainer	Action buttons
importActiveCancelButton	Cancel
importActiveSubmitButton	Final import

Recommended additions

Proposed data-testid	Reason
importActiveValidatingAlert	Assert validation loading without text-only selectors
importActiveValidationErrorAlert	Differentiate transient validation failure from file-list errors
importActiveViewErrorsButton / ExportErrorsButton	Open/export actions
importActiveErrorDialog / ErrorRow	Row, column, description assertions
importActiveFileRow / RemoveButton / RetryButton	Uploader state transitions
importActiveHistoryList / HistoryItem / HistoryLoader	Pagination and record assertions
importActiveHistoryEmptyState	Empty history

9. Product decisions needed before strict assertions

Decision	Question to lock down
Existing plate	Does final load always perform a full replacement, or only for specific asset states?
Blank optional cell	Canonical persisted value per field: null, empty string, zero, or preserve existing?
Mixed batch	Validation/load atomicity: reject all, partial success, or per-row result?
Headers	Must order be exact? Are additional columns rejected or ignored?
.xls	Is legacy BIFF actually supported by backend, or only advertised by frontend accept rules?
Mass limit	Maximum data rows and acceptable validation/load timing within the 10 MB cap.
History	Should successful import refresh history immediately and should failures show feedback?

Realistic stop condition: Do not automate guessed backend rules as flexible assertions. Mark those cases as contract-discovery tests, capture the actual response, agree the desired rule with product/backend, then make the assertion strict.

Appendix A. Current 102-column header contract

Exact names below come from the supplied workbook. Required/optional classification follows its header fill and comments. For maintainability, keep the canonical list in a version-controlled JSON/TypeScript fixture and review changes intentionally.

#	Col	Exact header	Req.	Workbook comment / constraint
1	A	ARTICULO	Required	Campo obligatorio Tipo: Número Máximo 20 caracteres
2	B	N° PLACA	Required	Campo obligatorio Tipo: Texto Máximo 20 caracteres
3	C	FOTOS (enlace)	Required	Campo obligatorio Tipo: Texto Máximo 200 caracteres
4	D	PLANILLA (ARCGIS)	Required	Campo obligatorio Tipo: Texto Máximo 100 caracteres
5	E	DEPARTAMENTO	Required	Campo obligatorio Tipo: Texto Máximo 4 caracteres Nota: Código DIVIPOLA
6	F	MUNICIPIO	Required	Campo obligatorio Tipo: Texto Máximo 5 caracteres Nota: Código DIVIPOLA - Dependiente de Departamento
7	G	LOCALIDAD	Optional	Tipo: Texto Máximo 50 caracteres Nota: Primera letra mayúscula
8	H	VEREDA	Optional	Tipo: Texto Máximo 50 caracteres Nota: Primera letra mayúscula
9	I	DOCUMENTO MEMORANDO SOLICITUD DE INGRESO	Optional	Tipo: Texto Máximo 25 caracteres
10	J	CONTRATO IPSE	Optional	Tipo: Texto Máximo 25 caracteres
11	K	CONTRATO CCG/FAZNI	Optional	Tipo: Texto Máximo 35 caracteres Nota: Condiciones generales del contrato
12	L	USUARIO DEL BIEN	Optional	Tipo: Texto Máximo 100 caracteres Nota: Nombres y Apellidos - Primera letra mayúscula
13	M	NIT/C.C. USUARIO	Optional	Tipo: Número Máximo 15 caracteres
14	N	PROYECTO	Optional	Tipo: Texto Máximo 15 caracteres Nota: ID del proyecto
15	O	S_E	Optional	Tipo: Texto Máximo 10 caracteres
16	P	CIRCUITO	Optional	Tipo: Texto Máximo 100 caracteres Nota: Primera letra mayúscula
17	Q	URBANO-RURAL APLICA PARA REDES DE DISTRIBUCION	Optional	Tipo: Texto Máximo 6 caracteres Nota: Aplica para redes de distribución
18	R	ALTITUD_COORD (ENLACE CARPETA	Optional	Tipo: Texto Máximo 100 caracteres
19	S	CONDUCTOR_TRENZADO	Optional	Tipo: Texto Máximo 10 caracteres
20	T	NODO ANTERIOR	Optional	Tipo: Texto Máximo 12 caracteres Nota: Identificación trazado de línea
21	U	NODO ACTUAL	Optional	Tipo: Texto Máximo 12 caracteres Nota: Identificación trazado de línea
22	V	DESCRIPCION CONCATENADA	Optional	Tipo: Texto Máximo 2000 caracteres
23	W	LATITUD_COORD	Optional	Tipo: Texto Máximo 30 caracteres
24	X	LONGITUD_COORD	Optional	Tipo: Texto Máximo 30 caracteres
25	Y	TIPO	Optional	Tipo: Texto Máximo 20 caracteres
26	Z	COD_LOCALIZACION_DANE	Required	Campo obligatorio Tipo: Número Máximo 6 caracteres
27	AA	NOMBRE_PLANILLA	Optional	Tipo: Texto Máximo 10 caracteres
28	AB	CONTRATO A.O.M	Optional	Tipo: Texto Máximo 10 caracteres
29	AC	OPERADOR A.O.M	Optional	Tipo: Texto Máximo 100 caracteres
30	AD	N° POLIZAS CONTRATOS DE OBRA	Optional	Tipo: Texto Máximo 30 caracteres
31	AE	FECHA_INICIAL_POLIZAS CONTRATO DE OBRA	Optional	Tipo: Fecha (DD/MM/AAAA) Máximo 10 caracteres
32	AF	OBSERVACION_ESTADO	Optional	Tipo: Texto Máximo 2000 caracteres
33	AG	FECHA_FINAL_POLIZAS CONTRATOS DE OBRA	Optional	Tipo: Fecha (DD/MM/AAAA) Máximo 10 caracteres

#	Col	Exact header	Req.	Workbook comment / constraint
34	AH	TIPO_DE_POLIZAS_CONTRATOS_DE_OBRA	Optional	Tipo: Texto Máximo 50 caracteres
35	AI	EDAD_TIPO_VIDA_UTIL	Optional	Tipo: Número Máximo 5 caracteres
36	AJ	EDAD_AGOTADA	Optional	Tipo: Número Máximo 5 caracteres
37	AK	VIDA_REMANENTE	Optional	Tipo: Número Máximo 5 caracteres
38	AL	UC_VALOR_T	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 16 caracteres
39	AM	AVALUO_RV	Optional	Tipo: Número Máximo 16 caracteres
40	AN	FUENTE	Optional	Tipo: Texto Máximo 100 caracteres
41	AO	FORMA DE ADQUISICION Y ORIGEN DE LOS RECURSOS	Optional	Tipo: Texto Máximo 16 caracteres
42	AP	PARTICIPACION DE PROPIEDAD DEL IPSE	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 4 caracteres
43	AQ	VALOR ASOCIADO A SU ADQUISICION	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 16 caracteres
44	AR	VALOR ASOCIADO A SU MANTENIMIENTO	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 16 caracteres
45	AS	MARCA	Optional	Tipo: Texto Máximo 30 caracteres
46	AT	CANTIDAD	Optional	Tipo: Número Máximo 5 caracteres
47	AU	CLASE	Optional	Tipo: Texto Máximo 30 caracteres
48	AV	SERIE	Optional	Tipo: Texto Máximo 30 caracteres
49	AW	CAPACIDAD(KVA)/(KW)	Optional	Tipo: Texto Máximo 5 caracteres
50	AX	FUNCIONAMIENTO EQUIPO	Optional	Tipo: Texto Máximo 5 caracteres
51	AY	RETIRADO_MTTO	Optional	Tipo: Texto Máximo 5 caracteres
52	AZ	ESTADO EQUIPO	Optional	Tipo: Texto Máximo 7 caracteres
53	BA	MUNICIPIO_LEV	Optional	Tipo: Texto Máximo 30 caracteres Nota: No requerido
54	BB	LOCALIDAD_LEV	Optional	Tipo: Texto Máximo 30 caracteres
55	BC	CODIGO_CREG	Optional	Tipo: Texto Máximo 15 caracteres
56	BD	Fecha levantamiento	Optional	Tipo: Fecha (DD/MM/AAAA) Máximo 10 caracteres Nota: No requerido
57	BE	PCBs	Optional	Tipo: Texto Máximo 30 caracteres
58	BF	TIPO DE AISLAMIENTO	Optional	Tipo: Texto Máximo 30 caracteres
59	BG	TIPO DE INSTALACIÓN	Optional	Tipo: Texto Máximo 8 caracteres
60	BH	EXISTE SISTEMA DE PUESTA A TIERRA	Optional	Tipo: Texto Máximo 5 caracteres
61	BI	CAPACIDAD (KVA)	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 5 caracteres
62	BJ	CAPACIDAD (KW)	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 5 caracteres
63	BK	LONGITUD CALCULADA	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 5 caracteres
64	BL	ALTURA (m) APOYO	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 5 caracteres
65	BM	DISPOSICION DEL APOYO	Optional	Tipo: Texto Máximo 10 caracteres
66	BN	TIPO MATERIAL APOYO	Optional	Tipo: Texto Máximo 20 caracteres
67	BO	RESISTENCIA MECANICA APOYO	Optional	Tipo: Texto Máximo 20 caracteres
68	BP	FTO APOYO OPERATIVO / IMPRODUCTIVO	Optional	Tipo: Texto Máximo 5 caracteres Nota: Operativo / Improductivo
69	BQ	ESTADO APOYO	Optional	Tipo: Texto Máximo 7 caracteres
70	BR	TIPO ESTRUCTURA: SUSPENSION/RETENCION	Optional	Tipo: Texto Máximo 9 caracteres Nota: Suspensión/Retención

#	Col	Exact header	Req.	Workbook comment / constraint
71	BS	FTO ESTRUCTURA OPERATIVO / IMPRODUCTIVO	Optional	Tipo: Texto Máximo 5 caracteres Nota: Operativo / Improductivo
72	BT	ESTADO ESTRUCTURA B / M	Optional	Tipo: Texto Máximo 7 caracteres
73	BU	No DE FASES	Optional	Tipo: Número Máximo 5 caracteres
74	BV	CALIBRE CONDUCTOR	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 5 caracteres
75	BW	TIPO MATERIAL CONDUCTOR	Optional	Tipo: Texto Máximo 20 caracteres
76	BX	CANTIDAD DE CIRCUITOS	Optional	Tipo: Número Máximo 5 caracteres
77	BY	FTO CONDUCTOR OPERATIVO /	Optional	Tipo: Texto Máximo 5 caracteres Nota: Operativo / Improductivo
78	BZ	ESTADO CONDUCTOR BUENO / MALO	Optional	Tipo: Texto Máximo 7 caracteres
79	CA	EDAD APARENTE (AÑOS)	Optional	Tipo: Número Máximo 5 caracteres Nota: Aplica a conductor/red
80	CB	CABLE GUARDA (M)	Optional	Tipo: Texto Máximo 20 caracteres
81	CC	PUESTA TIERRA	Optional	Tipo: Texto Máximo 5 caracteres
82	CD	TEMPLATESES	Optional	Tipo: Número Máximo 5 caracteres
83	CE	CANTIDAD CONDUCTOR (KM)	Optional	Tipo: Número Máximo 5 caracteres
84	CF	CODIGO CREG APOYO	Optional	Tipo: Texto Máximo 30 caracteres
85	CG	FECHA_FINAL_POLIZAS_CONTRATOS_DE_AOM	Optional	Tipo: Fecha (DD/MM/AAAA) Máximo 10 caracteres
86	CH	CODIGO CREG CABLE GUARDA	Optional	Tipo: Texto Máximo 30 caracteres
87	CI	CODIGO CREG PUESTA TIERRA	Optional	Tipo: Texto Máximo 30 caracteres
88	CJ	CODIGO CREG FIBRA OPTICA	Optional	Tipo: Texto Máximo 30 caracteres
89	CK	ATRIBUTOS APOYO	Optional	Tipo: Texto Máximo 500 caracteres
90	CL	VALOR CREG APOYO	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 16 caracteres
91	CM	ATRIBUTOS RED	Optional	Tipo: Texto Máximo 200 caracteres
92	CN	VALOR CREG CONDUCTOR	Optional	Tipo: Decimal Nota: Decimales con punto Máximo 16 caracteres Nota: (MT) x #mts x #fases
93	CO	N°_POLIZAS_CONTRATOS_DE_AOM	Optional	Tipo: Texto Máximo 30 caracteres
94	CP	FECHA_INICIAL_POLIZAS_CONTRATO_AOM	Optional	Tipo: Fecha (DD/MM/AAAA) Máximo 16 caracteres
95	CQ	TIPO_DE_POLIZAS_CONTRATOS_DE_AOM	Optional	Tipo: Texto Máximo 50 caracteres
96	CR	EMAIL_OPERADOR_AOM	Optional	Tipo: Texto Máximo 80 caracteres
97	CS	NO_CONTACTO_OPERADOR_AOM	Optional	Tipo: Texto Máximo 20 caracteres
98	CT	NOMBRE_CONTACTO_OPERADOR	Optional	Tipo: Texto Máximo 100 caracteres
99	CU	OPERADOR_RESPONSABLE	Optional	Tipo: Texto Máximo 100 caracteres
100	CV	FECHA_SUSCRIPCION_CONTRATO_AOM	Optional	Tipo: Fecha (DD/MM/AAAA) Máximo 10 caracteres
101	CW	VIGENCIA_DEL_CONTRATO_AOM	Optional	Tipo: Texto Máximo 20 caracteres
102	CX	CODIGO CREG CONDUCTOR	Optional	Tipo: Texto Máximo 30 caracteres

Appendix B. Source and implementation observations

Evidence	Location / observation
Import component	C:\Users\marango\Desktop\MfBienesElectricos\src\app\pages\actives\pages\import-active\import-active.component.ts/.html
Template	C:\Users\marango\Desktop\plantilla_importacion_2026-07-10_13.43.xlsx
Reference maps	Asset editing map, Create Asset robustness PDF, Import Active test-id PDF
Existing API pattern	tests/editAsset/apiPayload.spec.ts and tests/editAsset/apiResponse.spec.ts
Known frontend gaps	Placeholder alert title 'Prueba'; no visible template/history failure; history refresh only on tab selection; possible mojibake.